



AI technology in smart drying of foods: A critical review of research and applications

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ABSTRACT

Artificial intelligence (AI) and sensor technology will be the focus of future research on the intelligence of the food processing industry. The integration of AI and sensor technologies in the food drying industry can help visualize highly nonlinear, complex, and dynamically changing food drying processes to fully understand the drying kinetics and guide the optimization of the drying process. This review systematically summarizes the application of various AI technologies in food drying and presents several potential applications and limitations of various advanced AI combined with sensors (Electronic nose, Computer vision, Hyperspectral imaging etc.) for the real-time monitoring of food drying processes. In addition, according to the characteristics of different food drying processes, applicable AI technology and related sensor technology are summarized. Finally, the application and prospects of AI combined with 5G and IoT technologies for online real-time monitoring of food drying are discussed, with the aim of providing a new research direction in this field. Appropriate AI technology combined with sensor technology for different drying methods and food materials can process a large amount of characteristic data in real time, significantly reduce the calculation time and improve the prediction accuracy, optimize process parameters, improve the quality of dried food, and maximize energy saving. AI and sensor technologies have a bright future for the online real-time monitoring of food drying.

1. Introduction

In recent decades, the problem of food security has become increasingly acute as the population has increased dramatically and the demand for food has grown rapidly (Bara et al., 2024). A large amount of fresh food is wasted during transportation, storage, and sale because of spoilage. Therefore, drying is extensively used in food processing as an effective preservation method. Drying technology reduces water activity by reducing the water content of food materials to achieve freshness, which has the obvious advantages of reducing weight, reducing transportation costs, and extending shelf life (Wang et al., 2023). However, as progress in development continues in the food industry and the enhancement of consumer requirements for dry products, conventional drying methods such as hot-air drying (HAD), vacuum drying (VD), and freeze drying (FD) have made it difficult to satisfy the current needs of

food drying, owing to the longer drying cycle, poor sensory quality, poor retention of bioactive substances, and high energy consumption (Sun, Zhang, & Mujumdar, 2019). Some auxiliary technologies, such as the joint application of microwave, infrared (IR), radio frequency (RF), and other physical fields with drying technology, have emerged to accelerate the drying speed and enhance the drying quality of food, but there are also some problems (Jin et al., 2025). Cao et al. studied three drying methods (HAD, sun drying, and IR drying) of chili peppers and their effect on their drying characteristics and quality. The results indicated that IR drying significantly increased the drying rate and resulted in higher crispness values than HAD and sun drying, but both IR drying and HAD resulted in a higher degree of color loss (Z.-z. Cao et al., 2016). In addition, in another study, egg yolk powder prepared by FD and microwave-assisted FD was compared, and the quality of egg yolk powder prepared by MFD was similar to that of FD, and the drying

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period was shorter, but the drying end point was difficult to control, and it was prone to overheating (Wang et al., 2023). Because the drying system is difficult to control, the synergy of the physical field is often accompanied by uneven heating, structural damage due to the rapid migration of moisture, and excessive energy consumption, which negatively affects the flavor and nutrient content of food (Sun, Zhang, Mujumdar, & Yang, 2019; Wang et al., 2013). Therefore, the identification and control of moisture content (MC), drying endpoint and other parameters during the drying process are the focus of future research on intelligent drying (Jieling et al., 2020). However, the drying process of foodstuffs is usually a nonlinear, dynamic, multifactorial and complex process, and traditional mathematical modeling (empirical or semi-empirical models) usually requires a large amount of experimental data support and most of them can only achieve the expected results in specific cases (Sun, Zhang, & Yang, 2019). In addition, conventional modeling does not provide a good explanation for the mechanisms of chemical/biochemical reactions, structural and textural changes in the dried material, and compositional changes that may exist during the drying process.

Artificial intelligence (AI) is widely used to gain insight into food drying owing to its powerful arithmetic and learning capabilities. With increasingly deeper research on AI, the development of food drying systems based on real-time online monitoring has become possible (L. Cao, 2022). In addition, AI can provide decision-making services through learning from big data, which will overcome subjective consciousness bias, significantly reduce energy consumption, increase productivity, and provide strategies for sustainable development programs (Zeng et al., 2025). In recent years, many branches of AI have emerged, such as machine learning (ML) and deep learning (DL), which are already being used to solve practical problems (Tang et al., 2025; Y. Xu et al., 2023). Therefore, in the field of food drying, compared with traditional empirical models, AI provides a method for better modeling of complex and dynamic nonlinear drying processes. (Sun, Zhang, & Mujumdar, 2019). There have been several studies that utilize AI algorithms such as ANN and Extreme Learning Machines (ELM) to predict MC and other quality parameters for food drying (Asim Balbay et al., 2011; Wenchao et al., 2021; Wu et al., 2019). Previous studies showed that the ANN model (Root Mean Square Error = 9.0×10^{-4} , $R^2 = 0.994$) was significantly better than the semi-empirical model (Newton, Page, Two-term, Two-term exponential and Logarithmic model) in predicting the freeze-drying behavior of mushrooms (Tarafdar et al., 2019). In addition, the intelligent drying system relies on the integration of various types of advanced sensors in the drying system, in addition to the updating of algorithms to realize drying process visualization, automation, and intelligence (Ferreira et al., 2024). For example, electronic nose and machine vision in bionic technology and hyperspectral imaging technology can realize non-destructive capture of data such as visual, chemical/biochemical and morphological changes in the drying process, which in turn can realize intelligent analysis and automated control of drying parameters (R. Moscetti et al., 2021). These sensor technologies are fast, efficient, non-destructive, environmentally friendly, and sustainable, and are extensively utilized in the food processing and quality control industries.

At present, although various AI technologies have been discussed in relevant reviews, there are no relevant reports on the selection of AI models and their integration with sensor technologies. Thus, this study initially examines various AI technologies and their characteristics within the realm of food drying and then describes the role and potential applications of sensor technologies combined with AI for monitoring and optimizing the food drying process. Subsequently, this study explored the selection of appropriate AI models for different food classifications to solve unique problems. Finally, the future industrial landscape of food drying is envisioned and suggested, in combination with AI and 5G-IoT technologies. This review is expected to provide suggestions for future AI research in the direction of food drying and its application in the industry.

2. Application of AI technology in the food drying industry

With the deepening of research on AI technology, its potential value in the intelligence of food drying industry is inestimable. Therefore, it is important to understand the characteristics of different AI technologies for their effective application in the field of food drying (Table 1).

2.1. Machine learning

2.1.1. ANN

An artificial neural network (ANN) is a computational model inspired by the human brain and is composed of interconnected nodes (neurons) that are organized into layers (input, hidden, and output). ANN are designed with specific architectures, learning algorithms (supervised, unsupervised, and reinforced), and activation functions). Feedforward networks (Multi-layer Perception, Backpropagation, Radial Basis Function Network, Learning Vector Quantization) process information unidirectionally, whereas recurrent networks (Hopfield, Boltzmann, Kohonen) allow feedback loops. Learning rules, such as the delta rule (supervised) and Hebb's rule (unsupervised), adjust the connection weights to optimize the performance (Raj & Dash, 2022). ANN, often enhanced by optimization techniques (genetic algorithms, simulated annealing, and backpropagation algorithms), are widely applied to data modeling, classification, and optimization tasks. ANN is more suitable for hot air drying or spray drying of homogeneous foods (powder) to predict moisture content, drying rate and energy consumption. Currently, supervised learning algorithms are commonly used in the drying industry to decode potential patterns and relationships between input variables and drying result. Bannoud et al. used ANN to model the convective drying process of acai kernels, in which five sets of input quantities (air temperature and wind speed) were set, moisture content was measured as experimental data, and data separation (80 % for training, 20 % for testing) was used to study the effect on model training. The results showed that based on Root Mean Square Error (8.473×10^{-4}) and Correlation Coefficient (>0.9999), the ANN model exhibited superior drying kinetics modeling (Bannoud et al., 2024). Furthermore, Unsupervised learning (clustering and pattern recognition) reveals hidden patterns in drying data, whereas reinforcement learning optimizes drying strategies through real-time exploration of condition-product relationships (Gershman, 2016).

2.1.2. ANFIS

Compared to ANN, which is difficult to generalize because of its limitations in user interference and implicit meaning interpretation, the Adaptive Neuro-Fuzzy Inference System (ANFIS), as a composite computing model, integrates the strengths of both ANN and fuzzy logic and generates a more interpretable model using fuzzy reasoning methods in the ANN architecture (Chasiotis et al., 2021). The framework of ANFIS usually consists of a Sugeno fuzzy model (TSK) and a feed-forward neural network with learning algorithms, such as BP, which adjusts the parameters of TSK. ANFIS can integrate and efficiently model information from different sources including empirical models, heuristics, and data. A prototypical ANFIS model simulates a five-layer neural network architecture (Al-Mahasneh et al., 2016). Within this structure, the nodes of the first layer are adaptive and correspond to the input variables. The fixed nodes of the second layer are dedicated to the computation of membership functions for the input variables, whereas the neurons in the third layer represent the fuzzy logic rules, delineating the antecedents and consequences of the rules. The fourth layer comprises adaptive nodes, each of which executes a specific node function. The fifth layer consists of a single node that aggregates all input signals to produce the ultimate output. ANFIS is more suitable for intermittent drying or microwave drying of food materials with regular appearance (sliced fruits and vegetables) and can predict local moisture and drying stages. In an ANFIS case in food drying, Jumah and Mujumdar explored the use of ANFIS to model the intermittent drying process of corn.

Table 1
Comparative evaluation of various advanced artificial intelligence techniques in food drying.

AI technologies		Input data requirements	Complexity of training	Applicability of dynamic processes	Real time capability	Generalization of model	Reference
Machine learning	ANN	Structured time series or static data (temperature, humidity, wind speed, material thickness), need to be normalized.	Medium (sufficient samples were required to cover dry conditions, and the number of layers and neurons were adjusted).	The historical state cannot be retained, the temporality relies on manual construction of time Windows, and the spatial heterogeneity modeling is limited.	High (feedforward networks are computationally fast and suitable for embedded deployment).	Dependent on data quality and coverage, weak generalization to new drying conditions.	(LeCun et al., 2015)
	ANFIS	Structured data combined with fuzzy rules (e.g., membership function for “high temperature → fast drying rate”)	High (need to jointly optimize the neural network and fuzzy rules).	It relies on instantaneous input and manual adjustment of the rule base, which is less flexible.	Medium to high (fast inference speed, but the delay increases when the rule base is large).	The rule dependence is strong, and the generalization ability for new materials is weak.	(Chasiotis et al., 2021; Seyed-Hassan Miraei & Alex, 2024)
	PINN	Physical equations (such as Fick’s law of diffusion) and sensor data (temperature, humidity)	Extremely high (low data requirements, but coupling differential equations and neural networks, difficult training convergence).	Both temporal and spatial dynamics (e.g., modeling non-uniform water distribution).	Low (Solving physical equations is computationally intensive and requires GPU server participation).	Physical constraints enhance generalization and are suitable for extrapolation to new drying conditions (new materials, new equipment, etc.).	(M. Raissi et al., 2019)
	RNN	Long time series data are needed (e.g., hourly sampling, dry records lasting for weeks) and gradient disappearance/explosion needs to be dealt with.	High (large number of LSTM/GRU structure parameters, large amount of time series data required).	Temporal dynamic modeling (such as the transition prediction of drying stage) is optimal, while spatial modeling requires additional structure.	Low to medium (higher sequential inference latency, also requires GPU involvement).	The temporal patterns of similar drying processes generalize better.	(Mousumi et al., 2023)
	CNN	Large amount of image or grid data (such as temperature distribution of dry surface captured by infrared thermography, surface image during drying)	Low (Transfer learning for pre-trained models reduces complexity).	Strong spatial heterogeneity (such as local overheating region detection), temporal modeling needs to be combined with RNN (such as ConvLSTM).	Medium (requires GPU acceleration, single frame processing is fast, continuous video streaming delay is increased).	Strong generalization ability of image features, but weak generalization across sensors (affected by resolution).	(Miraei Ashtiani et al., 2021; Yilmaz, 2023)
Multimodal learning	GNN	Complete graph structure data (e.g., spatiotemporal relationships of all sensor nodes) is required.	High (graph convolution operation complexity is high, need to define adjacent matrix).	Spatial correlation dynamics applicable (e.g., multi-location drying rate interaction modeling).	Low (high computational resource demand for graph inference).	Weak generalization to graph structure changes (new sensor nodes need to be retrained).	(Deng et al., 2022)
	FL	Only a small amount of data is needed to verify the validity of the rules, relying on expert experience definition.	Low (only membership parameters need to be optimized when the rule base is fixed).	Poor dynamic adaptability (rules need to be manually adjusted), suitable for steady-state process control.	High (fast rule matching).	Fixed rules lead to poor generalization (new drying equipment needs to redefine rules)	(Adnan et al., 2015)
	Expert system	Symbolic input (e.g. “IF temperature > 55 °C AND humidity <25 % THEN increase wind speed”)	Very low (no training process, only relies on the rule base).	Suitable only for simple, deterministic dynamics (e.g., staged thermostatic control).	High (strong real-time rule matching).	Extremely weak generalization (unable to handle undefined combinations of conditions).	(Mujumdar, 2006)
Other technologies							

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Because it is difficult to simulate this multivariate, time-varying, and highly nonlinear interaction process with conventional mechanical modeling, researchers developed a MATLAB-based ANFIS model to simulate this holistic process, which consists of two sets of sub-models predicting the temperature and MC of solids. Finally, the overall performance of the model was evaluated in terms of the Root Mean Square Error (RMSE) and Correlation Coefficient (R^2) by training and testing the ANFIS using the MATLAB function “genfis1” (Jumah & Mujumdar, 2005). Although the introduction of the Type 2 fuzzy set architecture as an extension of ANFIS Type 1 addresses the uncontrollability of the Type 1 training parameters and structure and improves its prediction accuracy, the exponential growth in the number of fuzzy rules resulting from an increase in the input variables or affiliation functions still limits its ability to evolve. In addition, ANN and ANFIS suffer from a lack of memorization of the previous state, which limits their adaptive control capabilities (Seyed-Hassan Miraei & Alex, 2024).

2.1.3. PINN

Physics-Informed Neural Networks (PINN) harness the knowledge of natural phenomena to construct ML models. By integrating the physical principles encapsulated in partial differential equations and incorporating boundary and initial conditions into the loss function of artificial neural networks, PINN has emerged as a potent instrument for capturing the underlying dynamics of drying processes (Batuwatta-Gamage et al., 2022). They are adept at addressing complex problems characterized by high dimensions and intricate boundaries, making PINN particularly suited for simulating intricate interactions among multiple variables under ambiguous boundary conditions in food drying. Therefore, PINN is suitable for modeling complex mass and heat transfer drying (such as FD) of porous/heterogeneous foods (mushrooms, meat, etc.) to predict 3D water distribution, temperature gradient changes, and phase transition interfaces, among others. Compared with other AI technologies, PINN enhances interpretability and accuracy and remains effective even with limited training data. Manavi et al. integrated the mass and heat transfer parameters during the drying process into the PINN model architecture and validated the accuracy of the model predictions using the finite element method (FEM) and experimental data, demonstrating the high precision of the PINN model in capturing dynamic drying processes (Manavi et al., 2023). Consequently, PINN is advantageous for elucidating the latent physical phenomena within the drying processes.

2.2. Multimodal learning

As an advanced machine learning method, multimodal learning combines deep learning algorithms to improve the performance and accuracy of models through decision-level fusion by integrating data from different sources (images, sensor information, etc.). In the field of food drying, multimodal learning mainly includes recurrent neural network (RNN), convolutional neural network (CNN) and graph neural network (GNN).

2.2.1. RNN

To cope with the previously mentioned shortcomings of ANN and ANFIS, the recurrent neural network (RNN), as a DL algorithm, has memory that recognizes and captures timing data and integrates and feeds back information from the previous step by processing the input sequence. This mechanism gives the RNN the ability to memorize contextual information, and by organizing the input sequence, the RNN can simulate complex and variable dynamic drying processes and is more suitable for predicting drying dynamics in a temporal order (Mousumi et al., 2023). Liu et al. proposed a RNN (CS-OLS-RNN) using sparrow search and orthogonal least squares optimization to simulate microwave heating processes over time, which was verified by rice microwave drying experiments (T. Liu et al., 2020). In addition, several improved models, including Long Short-Term Memory (LSTM), Nonlinear Autoregressive Exogenous (NARX), and Elman Neural

Networks (ENN), have emerged to cope with the difficulty of RNN in modeling long-term dependent processes.

LSTM networks retain information for long periods by merging long-term memory units. This capability facilitates the modeling of long-term dependencies and recognition of temporal patterns during the drying treatment (Mousumi et al., 2023). The core principle of LSTM is its ability to selectively add or remove information from its internal cell state via gate mechanisms (input, forget, and output gates), which regulate the flow of information through the network, enabling LSTM to effectively simulate the complex dynamics of food drying.

NARX and LSTM architectures differ in how they handle temporal dependencies in input data. NARX networks use feedback connections to incorporate past target values and external inputs, propagating information from prior time steps to the current one. By leveraging historical data, including previous inputs and external factors, NARX networks enhance their predictive capabilities and effectively model the complex interactions and dependencies inherent in the drying dynamics. However, choosing appropriate network parameters, such as input and output embedding dimensions, the number of hidden neurons, and tap delays, presents a challenge for NARX networks. This limitation can impact both learning and generalization abilities.

ENN augments traditional neural network architectures by incorporating an echo state layer within the hidden layers to enhance memory retention. This echo state layer captures the outputs of the neurons in the first hidden layer and integrates them with the network's primary inputs, establishing a feedback connection between the first hidden layer and the input layer. This configuration enables the ENN to adapt to specific events and enhances the overall network stability. Compared with classical neural networks, ENN possesses superior computational capabilities and stability because of its internal feedback mechanism, allowing it to store and utilize output data from earlier time steps. This memory capacity enables the ENN to capture the temporal dynamics of drying processes and leverage drying conditions and historical information for learning and parameter adjustment, thereby optimizing the drying process and enhancing performance (Mousumi et al., 2023).

2.2.2. CNN

Convolutional neural network (CNN) is a DL architecture tailored for the processing of two- and three-dimensional signals, excelling in the analysis of image and video data. The fundamental strength of a CNN is its ability to autonomously extract hierarchical representations from raw input data. Through a cascade of convolutional layers, the network can identify salient features at various abstraction levels. This hierarchical feature extraction enables CNN to simultaneously capture fine-grained low-level details and comprehensive high-level semantic information, thereby enhancing the understanding of visual patterns (Miraei Ashtiani et al., 2021). Moreover, CNN rapidly simulates transport phenomena and even substitutes partial differential equations for such phenomena through generative adversarial networks or by employing physics-informed loss functions for weakly supervised learning. Even under ambiguous boundary conditions, CNN can construct robust models to address issues related to heat conduction and gas flow (Maziar Raissi et al., 2020). Furthermore, the integration with sensor technology amplifies the advantages of DL algorithms, enabling the effortless extraction of feature data from complex drying processes via sensors, thereby streamlining the research process (Yang et al., 2024). Therefore, CNN is suitable for the dynamic prediction of drying process of food with obvious appearance characteristics. You et al. monitored the drying process of Tencha using a CNN combined with computer vision to predict water distribution (You et al., 2024). This approach is well suited for the targeted handling of real-world data susceptible to noise, errors, and less-than-ideal conditions. The robust image processing and learning capabilities of DL significantly abbreviate model training and inference procedures, holding immense potential in the field of food drying applications.

2.2.3. GNN

As a new DL model, GNN performs well in graph structure data processing. Although the relevant research in the field of food drying is limited, GNN's outstanding performance in the analysis of molecular structure makes it have great potential to analyze the complex chemical changes of food drying at the molecular level (Wieder et al., 2020). In the drying process, the protein, oil and starch in the food will undergo significant physical and chemical changes during the drying process and even interact. These changes not only affect the texture, taste and flavor of the food, but may also affect its nutritional content and shelf life. GNN can describe the dynamic structural changes of molecules through the evolution of graphs through the message passing mechanism, and extract features from them to predict the influence of key parameters in the drying process on molecular changes, thus helping researchers to understand the drying process deeply and providing guidance for optimizing the drying process. In the field of biomedicine, Reau et al. have developed a GNN-based deep learning framework for deep understanding of the interactions between protein molecules (Reau et al., 2023).

2.3. Other AI technologies

2.3.1. Fuzzy logic

Fuzzy logic (FL), an AI technique emulating human reasoning and decision making, comprises fuzzy rule bases, fuzzification, fuzzy reasoning, and defuzzification (Qin & Litchfield, 1993). Expert knowledge is encoded into the IF-THEN rules to form a rule base. Precise inputs are fuzzified, processed via fuzzy inference using the rule base to yield a fuzzy conclusion, which is then defuzzified into a crisp output (Adnan et al., 2015). Applications include proportional-integral-derivative (PID) controllers, intelligent monitoring, and parameter optimization. Fuzzification involves defining fuzzy sets (Zadeh representation) and membership functions (triangular, trapezoidal, S-shaped, and Gaussian) (Dai et al., 2017). Fuzzy inference employs max-min compositional rules, whereas defuzzification methods include the centroid, mean of maxima, and various maximum-based approaches. Based on the advantage of effectively dealing with various uncertainty and ambiguity factors existing in the drying process, the application of fuzzy logic in food drying is mainly embodied in the control and optimization of the drying process, such as dynamically adjusting the control of temperature and humidity, drying time, and predicting the drying curves and drying quality according to the type of food products, initial MC, target MC, and other factors. In addition, FL is often used in conjunction with experts or fuzzy controllers.

2.3.2. Expert system

Expert systems that employ knowledge representation and reasoning to mimic expert problem solving are branches of AI. They are categorized by knowledge representation (logic-based, rule-based, semantic network, frame-based) and application (e.g., diagnostic, predictive, control) (Linkens & Chen, 1995). A typical expert system comprises a human-machine interface for user interaction, a knowledge acquisition module (a key challenge in development), a knowledge base storing expert knowledge (crucial for system performance), an inference engine for rule-based reasoning (using IF-THEN rules with AND, OR, NOT logic), a dynamic database for temporary data storage, and an interpreter for presenting results. The knowledge base, which is independent of other system components, allows for modification and improvement. In the field of drying, expert systems are used to optimize drying parameters, quality control, process simulation and optimization, and food safety control, and to perform the resultant data analysis to guide business decisions. Therefore, for better control of the food drying process, a comprehensive food material database is essential, which should contain the MC, chemical composition, and structural and thermodynamic attributes of food materials, and, of course, suggestions for the appropriate drying method should be included. At present, there

have been studies on expert systems based on FL for evaluating quality changes in peptide and protein FD processes (Trnka et al., 2013). In addition, to make the system better understand natural language, expert system based on large language models (LLMs) is worth considering developing. LLMs can understand the user's simple description and extract relevant knowledge from the expert system to provide professional recommendations and solutions. In addition, thanks to the automated information extraction capabilities of LLMs, it is possible to automatically analyze the latest research literature and industry reports and regularly update the knowledge base of the expert system.

3. Combined application of AI-based sensor technology in food drying

As an excellent detection tool, sensor technology is widely used in the food field. It is well known that training AI models requires a large amount of real-time data, which cannot be met by traditional measurement methods, and sensor technology can largely solve this problem. In addition, the fusion of sensor technology and AI data can not only obtain a large amount of real-time data without damage but also process and model the data in a very short time, effectively saving time and cost.

3.1. Biomimetic technology

In recent years, with the advancement of artificial intelligence (AI) in the field of food drying, sensor-based biomimetic technologies (electronic nose, electronic mucosa, and computer vision) have garnered significant attention because of their real-time capabilities, object-friendliness, efficiency, and versatility (Fig. 1A, B, C). Empowered by ML and DL, biomimicry is emerging as a superior technique for simulating food-drying processes and subsequent evaluation procedures (Su et al., 2015).

3.1.1. Electronic nose and electronic mucosa

Monitoring the sensory attributes is a crucial aspect of the drying process. Improper techniques during drying that induce unnecessary physicochemical changes in food can result in unpleasant sensory properties of the dried products. Loss of volatile aromas and the generation of off flavors are the primary reasons for the poor sensory characteristics of dried foods, which may lead consumers to question their nutritional and functional attributes. Hence, it is essential to prevent evaporation of the original aroma components in food during drying to maintain market competitiveness. Since sensory attributes (aroma volatilization) are real-time, ongoing processes during drying, traditional methods of conducting real-time aroma measurements on food are highly challenging in terms of precision and efficiency, and are often not cost-effective (Ghasemi-Varnamkhasti & Aghbashlo, 2014). Moreover, owing to the drying process involving mass and heat transfer, dynamic nonlinearity, and potential physicochemical and biochemical reactions between different food components, traditional empirical models of offline detection no longer meet the current requirements. Consequently, the integration of biosensors, such as electronic nose and electronic mucosa, with drying machinery, providing an integrated system for internal real-time monitoring, detection, analysis, and feedback control, has been increasingly adopted and applied by the food industry (Ghasemi-Varnamkhasti, et al., 2014). The electronic nose (E-nose) is a device inspired by human olfaction designed to identify gases and odors by simulating the structure of the human nose. It typically comprises a gas delivery system and analytical sensors that respond to various chemical substances to recognize specific odor patterns, ultimately determining the composition or quality of the sample (J.-C. Xu et al., 2017). E-noses are increasingly seen as the preferred method for real-time sensory attribute measurement of processed foods, providing reliable, non-invasive, and non-destructive assessments. They are also used to develop monitoring and control strategies for food drying

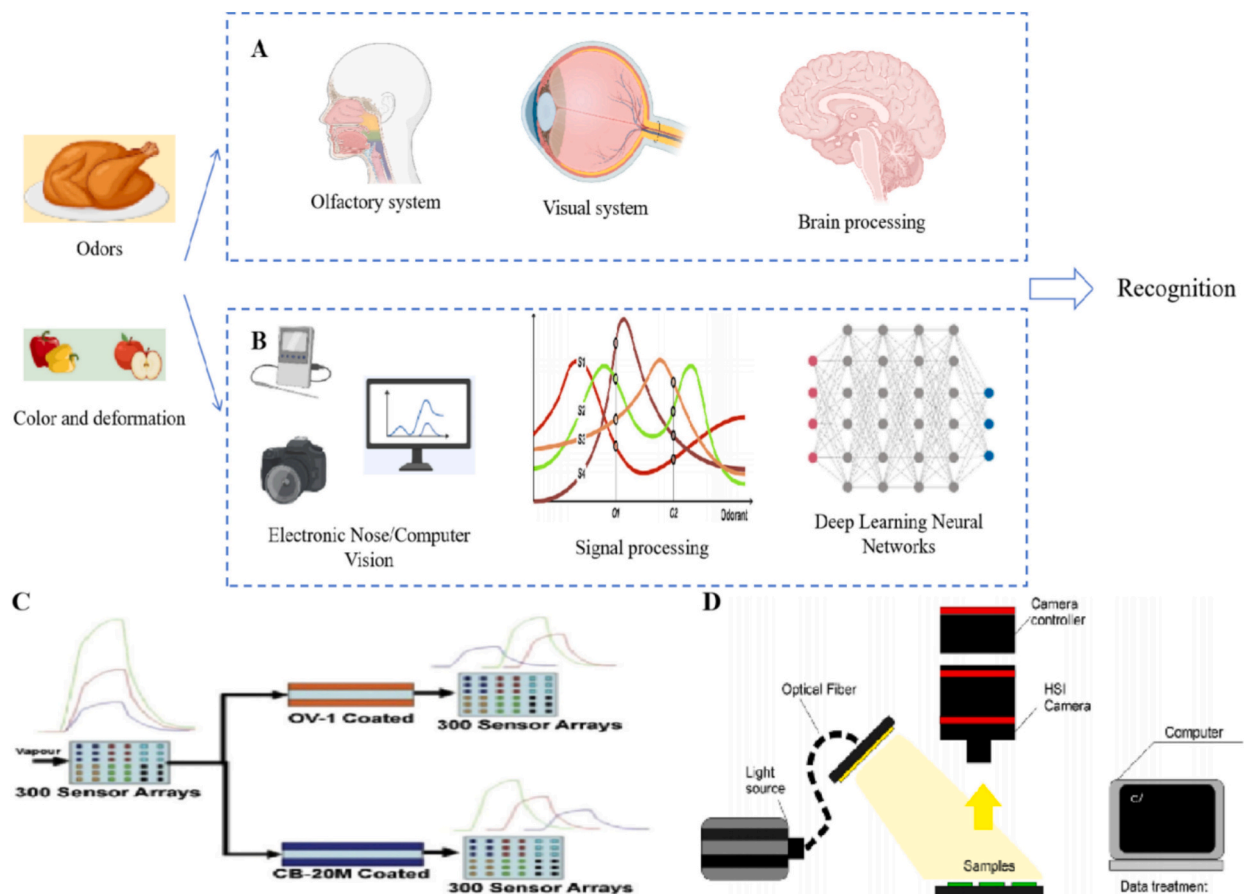


Fig. 1. Comparison of biological and AI-based olfactory/visual systems (A and B); Schematic diagram of an electronic mucosa (C); The basic structure of HSI system (D) (Ferreira et al., 2024; Ghasemi-Varnamkhasti, et al., 2014).

processes in a nonintrusive manner. Fei et al. employed GC-MS and E-nose to detect changes in volatile components of sliced mushrooms during the freeze-drying (FD) process and found that the results from the E-nose were largely consistent with those obtained from GC-MS (Fei et al., 2016). Compared to traditional gas chromatography and gas chromatography-mass spectrometry, the E-nose offers significant advantages as an innovative instrument, including reduced sample volume, high sensitivity, faster analysis speed, smaller size, and lower cost (Buratti et al., 2006). Another important benefit of E-nose is its integration with AI. In the food drying processes, the volatile aroma of food is captured by the E-nose, which is then processed into raw aroma profiles and subjected to signal processing, followed by filtering and confirmation through ANN or expert systems. This approach not only simplifies experimental procedures compared to traditional laboratory methods but also reduces various errors that are prone to occur in manual experiments. Although there are currently no cases of E-nose being used for the online monitoring of food drying, several AI-driven E-nose systems for food detection have emerged. Zhang et al. proposed a low-cost ML-driven E-nose odor detection system that utilizes a support vector machine (SVM) classification algorithm to distinguish between various fruits and vegetables, achieving an accuracy of 96.7 % (Zhang et al., 2024). Additionally, Lee et al. combined E-nose with ML, employing a density-based spatial clustering of applications with noise (DBSCAN) algorithm to accurately identify coffee adulteration (Lee & Rianto, 2024). Future research will focus on algorithm-driven E-nose systems for monitoring various food drying processes, as E-nose is compatible with any drying apparatus and facilitates smart automation of food drying facilities. However, the E-nose still exhibits certain limitations; its sensitivity and detection capabilities can be adversely affected by temperature, pressure, and background noise, particularly in

drying methods such as FD or spray drying (SD), which are prone to sensor interference. Additionally, the difficulty of sensors in identifying specific types of compounds may lead to ineffective recognition by the E-nose (Roeck et al., 2008). To address these challenges, several studies have been initiated: replacing conventional sensors with high-sensitivity acoustic wave sensors or gas sensors based on nanowires to reduce response times; optimizing ML models to decrease data training durations and thereby lower costs; and implementing active sampling control and online calibration to enhance long-term reliability. As our understanding of human olfaction has deepened, more sophisticated biomimetic technologies have emerged. The electronic mucosa (EM), an advanced version of the E-nose, consists of a distributed sensor array and artificial mucosa (Harun et al., 2012). This biosensor is equipped with varying amounts of binding proteins, enhancing its performance compared to the traditional E-nose, offering higher selectivity, greater sensitivity, and faster response times. When different gases enter the electronic mucosal system, they interact with an absorption coating analogous to the stationary phase in gas chromatography, resulting in varied transit speeds and retention times. Harun et al. developed a portable electronic mucosa (PeM) system that integrates three large sensor arrays, each containing 576 sensors and featuring 24 different coatings to improve selectivity for unknown odors (Harun et al., 2012). The PeM system captures both spatial and temporal information. Additionally, Gardner and Taylor employed a novel DL (convolutional) algorithm for signal processing to address the robustness of sensor drift and noise encountered in practical applications (Gardner & Taylor, 2009). Although there is currently no related research in the field of food drying, EM holds significant potential as a real-time online monitoring system for quality and safety control in food drying processes.

3.1.2. Computer vision

Drying induces transformations in the sensory profiles of foodstuffs, and beyond the modulation of flavor during the dehydration process, visual attributes such as morphology, dimensions, hue, and textural characteristics of dehydrated foods are pivotal in determining consumer acceptance. Hence, real-time surveillance of superficial attributes during food dehydration constitutes a formidable challenge. Conventional methodologies are inadequate to satisfy this requirement, whereas computer vision (CV) technology, as a noninvasive, expedient, and precise technique, can evaluate these metrics *in situ* during the dehydration procedure, thereby augmenting the quality of dehydrated food merchandise (Aghbashlo et al., 2014). Computer vision can be used to detect shrinkage, chromatic alterations, and textural modifications in food substrates undergoing desiccation. By leveraging these surveillance data, the dehydration conditions can be modified expeditiously to guarantee the integrity of the ultimate product. A computer vision system is fundamentally structured into five phases: 1) data procurement through an array of image sensors. 2) Preliminary processing of the procured images encompassing deblurring, compression, noise attenuation, and contrast amplification. 3) Division of images into discrete segments. 4) Institution of thresholds for delineation of image features. 5) Elaborate processing of delineated images, entailing categorization and dissection of targeted parameters. Owing to the desiccation and thermal stress encountered during food dehydration, the morphology, dimensions, coloration, and textural attributes of comestibles commonly undergo transformation.

Deformation is principally caused by structural alterations, including changes in density, porosity, and specific volume, which are a consequence of moisture translocation and fluctuations in vapor pressure within the food matrix during the drying phase (Hosseinpour et al., 2011; Koc et al., 2008). Of particular significance is the fact that marked deformation can seriously compromise the sensory characteristics and rehydration potential of dehydrated commodities, detracting from their visual allure to consumers and potentially precipitating the failure of the entire dehydration endeavor (Aghbashlo et al., 2010; Fan et al., 2005). Such setbacks entail an escalation in futile energy utilization and an enlargement of the carbon footprint. Some general concepts have been considered by researchers, and the measurement and estimation of deformation during food drying using traditional methods and models have been thoroughly studied (Krokida & Maroulis, 1997). Nevertheless, the lag of this offline mode, the limitations of traditional measurement methods, and the lag of manual experiments can affect the original drying mechanism of the food and lead to significant errors. CV provides an effective approach for the online analysis of the effects of drying on food morphology (Campos-Mendiola et al., 2007; Yadollahinia & Jahangiri, 2009). Chakravartula et al. developed a monitoring system using CV technology, analyzed the morphological changes of carrot slices during convective drying through image acquisition, and found that their shrinkage was linearly correlated with water loss (Nallan Chakravartula et al., 2023). CV typically requires large amounts of data to predict food drying processes, yet traditional empirical models may struggle to process large amounts of data quickly. Thibault et al. used AI models such as ELM, SVM and evolutionary polynomial regression (EPR) to predict the porosity of food after drying, and the predicted results were highly consistent with the experimental results (Bruno et al., 2024). In addition, Khazaei et al. developed an ANN-based prediction model combined with CV to predict the shrinkage during grape drying. This study shows that the artificial neural network model has better predictive performance than the traditional multiple linear regression (MLR) model (Behroozi Khazaei et al., 2013). These studies showed that AI-enabled computer vision can monitor and analyze information such as shape, size, and surface changes during food drying in real time and provide acceptable and relevant analysis results.

Product color is a primary factor influencing consumer acceptance and preference, as it directly stimulates visual perception (Wang et al., 2023). Moreover, color is a significant factor affecting drying quality

because of its ability to reflect the chemical and biochemical reactions occurring during food drying, and indirectly indicates the loss of bioactive compounds, thereby revealing the nutritional and functional properties of dried foods (Yuanyuan et al., 2022; Yue et al., 2025). These reactions are profoundly influenced by the drying techniques and conditions. In fruit and vegetable drying, the initial color can signify the freshness of the product, whereas inappropriate color changes induced by drying conditions, such as darkening caused by excessive temperatures, can result in poor market reception. These abnormal color changes may be attributed to the degradation of certain active substances, such as chlorophyll and anthocyanins (Nguyen et al., 2023). Consequently, the ability of dried foods to maintain minimal deviation from their initial color is indicative of the success of the drying process. Traditional offline measurement methods primarily involve comparing the color of dried food with its initial color through visual inspection or colorimetry. However, such simplistic approaches fail to monitor dynamic color changes during the food drying process, rendering them unsuitable for most food industries because of the associated risks and irreversible nature, which can lead to resource wastage. Although manual measurements at regular intervals can be facilitated by the convenience of laboratory equipment, this method is often accompanied by significant errors owing to sampling and preparation, particularly for air-sensitive foods. Additionally, these methods typically exhibit a degree of latency, which limits the possibility of real-time adjustment of experimental parameters. With the application of low-cost computer vision (CV) technology in the drying field, non-destructive, rapid, and real-time measurement methods have been established that are suitable for continuous monitoring of the drying process (Rosas-Romero et al., 2022). Li et al. (Li et al., 2024) utilized CV combined with image recognition technology to analyze the correlation between color changes and anthocyanin degradation in black carrots during air-impingement jet drying (AIJD). Furthermore, certain optical enhancement techniques have been employed to augment the precision of computer vision (CV) technology, such as optical scattering and laser diode backscatter (Udomkun et al., 2017), among others. Onwude et al. (Onwude et al., 2018) developed a neural network model based on CV and optical backscatter imaging to monitor the shrinkage and color changes of sweet potatoes during drying. This model employs a multilayer perception (MLP) with a three-unit input layer, two hidden layers consisting of 18 neurons each, and a five-unit output layer. The results indicate that the CV can accurately provide dynamic color information in both offline and real-time modes. Owing to advancements in AI technology, CV systems based on DL have significant potential for enhancing the understanding of biochemical reaction mechanisms and color-change dynamics during the drying process.

3.2. Hyperspectral techniques

Although AI integrated with CV technology has been extensively researched for monitoring food color, deformation, and predicting MC, it is limited by the challenge of directly acquiring information on internal structure and compositional changes during food drying. These parameters typically require destructive measurements in the offline mode, such as texture analyzers. However, this method introduces latency and potential errors that are associated with destructive testing. In recent years, with the emergence of intelligent drying systems, non-destructive techniques such as near-infrared spectroscopy (NIRS), Fourier transform infrared spectroscopy (FTIR), and hyperspectral imaging (HSI) have successfully enabled the real-time acquisition of texture and compositional changes during food drying processes.

3.2.1. NIR and FTIR

NIR spectroscopy utilizes molecular vibrations to absorb electromagnetic radiation, and these absorptions reflect changes in the internal composition of a sample. NIRS technology is extensively employed for the nondestructive detection of critical parameters such as MC and

chemical composition during food drying processes. For instance, in the drying of apple slices, the combination of NIRS technology with Principal Component Analysis (PCA) can rapidly and accurately evaluate the quality indicators of dried apple slices. However, the NIR results may be subject to background noise or overlapping phenomena due to environmental or sample factors. ML algorithms, such as PCA and chemometric methods, can precisely pinpoint specific components that are reflected in the infrared spectra through processes such as screening and noise reduction. Owing to its high sensitivity to water molecules, NIR is commonly used to reflect drying rates and predict MC. Sheng et al. developed a real-time model for predicting MC in dried black tea by integrating near-infrared spectroscopy with computer vision (CV) and employing a data fusion strategy. The results indicate that the model based on intermediate-level data fusion achieved a correlation coefficient (R_p) of 0.9696 and RMSE of prediction of 0.0016 for the prediction set, with an RPD of 4.0846 (Sheng et al., 2023). Additionally, researchers have employed NIRS technology in conjunction with chemometric methods to establish predictive models for estimating MC and other quality parameters (such as color and texture) of carrots at different stages of HAD (Roberto Moschetti et al., 2017). The results demonstrated the accuracy of NIRS for real-time monitoring of MC and other quality attributes of carrots during HAD, which aids in the optimization of drying processes. However, this study also highlights the need for improved model generalization capabilities. In addition, FTIR, akin to NIR, is a spectroscopic technique that employs Fourier transform

technology to analyze the infrared absorption properties of samples for molecular structure identification, with its measurement wavelength typically spanning from 4000 to 400 cm^{-1} (Voort, 1992). FTIR is increasingly employed in real-time food applications to predict compositional changes as a nondestructive technique, particularly in the realm of foods rich in bioactive substances. He et al. (Gang et al., 2023) utilized FTIR in conjunction with PCA to identify the drying process of Amomum tsao-ko and developed discriminant models using partial least squares discriminant analysis, SVM, and residual neural networks (ResNet). The SVM and ResNet models achieved prediction accuracies exceeding 99 % for the temperature. Liu et al. (X. Liu et al., 2024) have developed a rapid prediction model using ML-assisted FTIR to forecast freeze-drying (FD) parameters. Among the 18 selected models, ANN exhibited superior performance during the pre-freezing and primary drying stages, whereas the random forest model was more suitable for the sublimation phase, with a prediction rate as high as 96 %. In addition, a portable FTIR system was introduced for the first time for online monitoring of industrial food and bioprocessing applications (Bijay et al., 2024).

3.2.2. HSI

Hyperspectral imaging (HSI) technology, an integration of spectroscopy and imaging, has tremendous potential for non-destructive testing in the food industry (Fig. 2D). HSI typically operates in three modes, each tailored for different sample types: 1) the reflectance mode

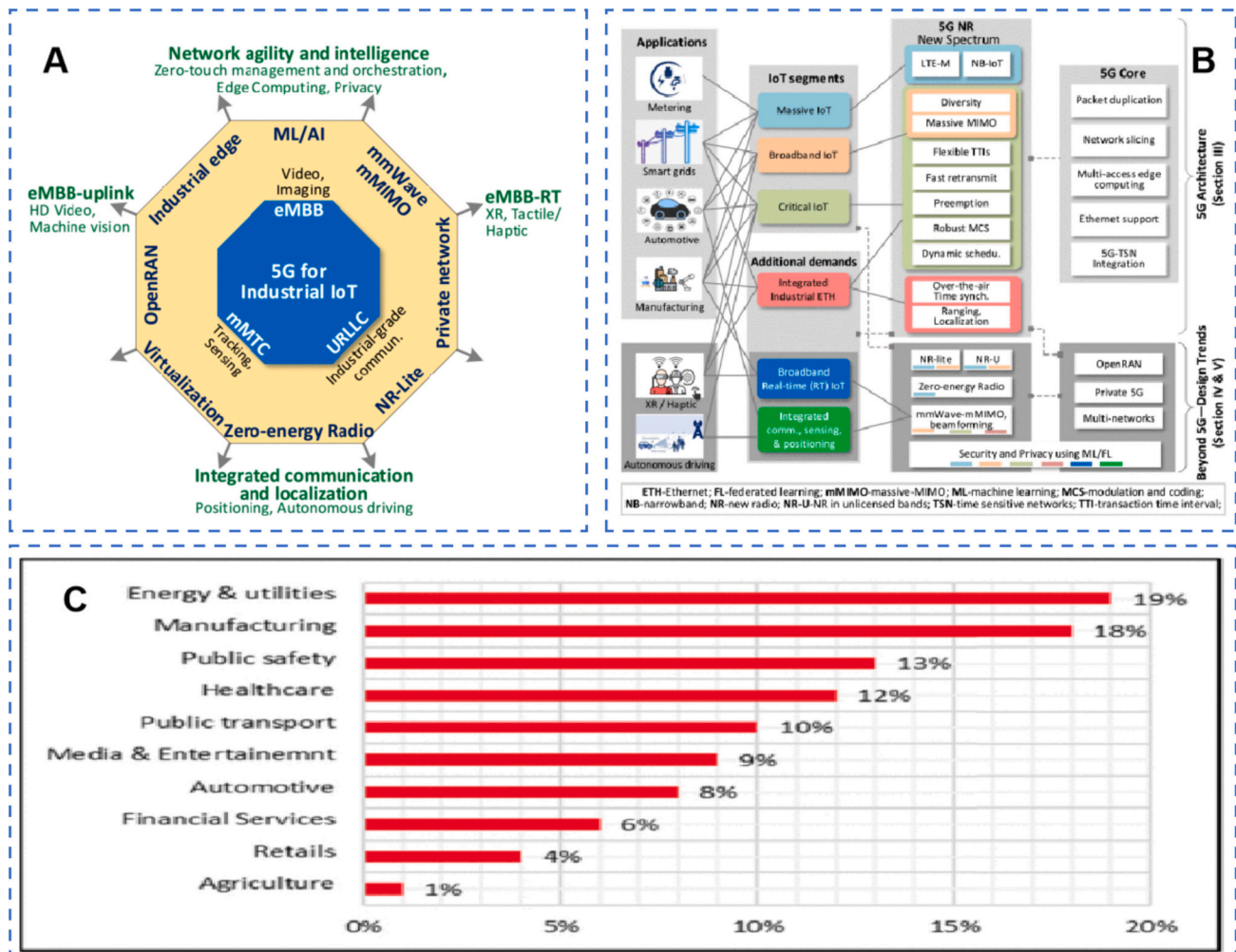


Fig. 2. Industrial IoT architecture and design trends based on AI and 5G (A) (B); Projected gross revenues in 2026 for related industries supporting 5G (C) (Humayun et al., 2020; Mahmood et al., 2022).

is commonly utilized to characterize the external attributes of samples; 2) the transmission mode offers insights into the changes associated with internal compositional variations; and 3) the scattering mode, amalgamating the advantages of the former two, is suitable for the examination of thicker samples such as fruits and vegetables (Nicolai et al., 2014). Furthermore, the advent of HSI has addressed the limitations of traditional methods for assessing MC, such as prolonged analysis time, destructiveness, and inability to determine moisture distribution at a unit sample level. HSI allows for real-time monitoring of color changes and other quality indicators during the drying process of food products. Ndisya et al. (Ndisya et al., 2021) developed a predictive model based on HSI and chemometric methods for the hot air-drying process of purple-flecked taro slices. This model was designed to monitor crucial parameters, such as MC, color, chemical composition, and structural properties, ultimately yielding excellent prediction results ($RMSEP < 2.0$, $r_p^2 > 0.90$, $RPD_p > 3.5$). Lin et al. developed a moisture distribution model for microwave vacuum drying (MVD) of different vegetables using the HSI system. The study employed various algorithms, including Partial Least Squares Regression (PLSR), LS-SVM, and Genetic Algorithm-MLR for the

visualization of MC (Lin & Sun, 2022). In a study on shrimp HAD, PLSR and LS-SVM models were established to monitor moisture distribution via low-field nuclear magnetic resonance (NMR) technology and analyze its correlation with color and texture attributes. The results indicated that, for MC (MC), the LS-SVM model based on full-band spectral data exhibited the best performance, with an RPD value of 2.814. For the color and texture parameters, the LSSVM model yielded optimal results, with RPD values of 3.292, 2.753, 3.211, 2.807, and 2.842, respectively (W. Xu et al., 2022). In addition to common spectral imaging techniques, terahertz time-domain spectroscopy has great potential for predicting food dryness owing to its strong penetration, real-time, and moisture sensitivity. Ren et al. (Ren et al., 2024) integrated a THz-TDS imaging system with fluidized bed drying technology to combine image data and datasets using CNN and long short-term memory (LSTM) for the MC loss prediction of zanthoxylum and sweet corn blends. The aforementioned studies demonstrated that the combination of HSI with ML can effectively predict key parameters during food drying processes, such as MC, color changes, and internal characteristics (Lianfei & Renshuai, 2024). This approach not only yields more

Table 2

AI models and sensor technologies are used in different classifications of food drying.

Products	AI-Sensor technologies	Aims	Results	Reference
Fruit and vegetable drying	Kiwifruit	GA combined with FL, CV	Optimize the drying process and monitor changes in color and shape.	The system significantly reduces drying cycles and improves appearance properties. (Hossein Nadian et al., 2017)
	Cantaloupe	ANN, ELM, CV	To model and predict the MC based on image analysis.	Compared with ANN, ELM shows superior performance, shorter training time and better anti-overfitting characteristics. (G. Zhu et al., 2023)
	Apple	ANN, K-nearest neighbor, Random Forest, GP, SVR	To assess total phenol, antioxidant and ascorbic acid content.	SVR and GP had the best results. (Cetin & Saglam, 2022)
	Red dates	CNN, BiLSTM, SE	To monitor changes in soluble solids content (SSC), titratable acidity (TA), MC, and hardness during HAD.	It can directly reflect the temporal and spatial distribution of quality attribute parameters of red jujube at different drying stages and has a good prediction effect. (Q. Liu et al., 2025)
Meat drying	Dry-cured ham	CNN, CV	To assess intramuscular fat (IMF) content.	CNN can correctly segment IMF in dry-cured ham, and its performance is like that of traditional images, but it is limited by image quality. (Munoz et al., 2019)
	Dried cured meat	E-nose, DFA	To conduct non-destructive detection of penicillium strains of ochratoxin in dried meat.	The recognition rate was 73 %. (Lippolis et al., 2016)
	Raw pork tripe	ANN	To predict mass transfer kinetics and mass change.	The model mathematically correlates the process command variables (temperature, sugar and liquid smoke flavoring concentrations of the immersion solution) with the end-product characteristics (water loss, salt, phenol and sugar gains, water activity and pH patterns). (Poligné et al., 2002)
	Chicken breast fillets	ANN	To establish a prediction model for the drying characteristics, rehydration, thermodynamic and spectral characteristics of chicken breast slices.	ANN model has better prediction effect than semi-empirical model. (Deepak et al., 2019)
Nut drying	Macadamia nuts	HSI, ML (PLSR, ANN, SVM, GRP)	To predict the MC of individual macadamia nuts.	PLSR has the highest prediction accuracy for shelled nuts and kernels in 760–967 nm band, while ANN and GRP have the highest prediction accuracy for cores. (Farrar et al., 2024)
	Canarium and Macadamia nuts	HSI, ANN, PLSR	To predict peroxide value (PV), free fatty acid (FFA) level and fatty acid concentration.	The prediction accuracy was higher, but the prediction accuracy of PV and linoleic acid was worse. (Gama et al., 2024)
	Walnut	GA-ANN	To model drying time (DT), specific energy consumption (SEC), high grain rate (HR), total grain rate (WR), shell breaking rate (SR) and other parameters during the drying process.	The optimum process parameters (infrared temperature, wind speed, MC, loading direction) are obtained, and the relative error is less than 4 %. (Taoqing et al., 2023)
	Biltmore nuts	ELM	To model the drying process of fixed bed.	The ELM model can better capture the trend of drying rate over time and consider the influence of multiple factors (external environment). However, it is found that the forecasting ability of ELM under long-term forecasting or extreme conditions needs to be further verified, and the generalization ability of the model needs to be strengthened. (Balbay et al., 2012)

precise and real-time results but is also non-intrusive, preserving the integrity of the samples (Ebrahim et al., 2023). Moreover, combined with AI, it can be integrated into intelligent dryers, predictive modeling, and process automation, thereby enhancing industrial drying practices. However, HSI systems are typically characterized by high costs and significant computational requirements. Future research should focus on simplifying these systems and integrating them into practical production environments.

4. Application of AI in drying of different classified food products

The application of AI technology in the field of food drying has great potential, and various AI models have emerged (such as ANN, SVM, RNN, and GBM) for such applications. However, different foods have different characteristics, and the focus of attention on the drying process is also different. Therefore, to maximize production efficiency, reduce energy consumption and costs, and maintain product quality and safety in combination with consumer preferences, it is essential to select an appropriate AI for the drying of different types of food (Table 2). When selecting a model, it is essential to consider the data type, amount of data, prediction target, complexity, and interpretability. In general, data preprocessing, feature engineering, and model tuning are the key steps in the successful application of AI models.

4.1. Fruit and vegetable drying

The demand for fruits and vegetables has been increasing in recent years due to their obvious health benefits (Azoubel et al., 2015). However, due to the limitations of time and space, there is often 40–50 % waste and huge costs in the process from harvesting and storage to transportation, sales and processing (Porat et al., 2018). As an effective fresh-keeping method, drying technology is widely used in fruit and vegetable processing (Min et al., 2017; Nascimento et al., 2016). Fruits and vegetables, characterized by their high MC, often face extended drying cycles when employing traditional drying techniques, leading to issues such as appearance deterioration, excessive nutrient loss, and high-energy consumption. The visual attributes of dried fruits and vegetables are pivotal in influencing consumer preference and market sales, encompassing the preservation of original color, aroma, and shape (Mierzwa et al., 2019). Consequently, the control and optimization of the drying process is critical to the edible and nutritional value of these food products. Previous research has primarily focused on optimizing drying parameters through extensive experimental studies or enhancing the pretreatment and drying process in conjunction with physical field optimization to improve product quality and production efficiency (Chaowalit et al., 2019). Efficient physical field technologies, such as microwave, radiofrequency, infrared radiation, and ultrasound, are widely used to assist in the drying of fruits and vegetables to reduce drying cycles and energy consumption and to improve drying quality (Jieling et al., 2020). However, this often necessitates substantial manual intervention and is subject to time lags, making it challenging to achieve desired outcomes in actual production. AI has been extensively employed to enhance the monitoring of fruit and vegetable drying processes to control production quality. Nadian et al. developed a genetic algorithm (GA)-based ANN combined with an intelligent fuzzy control system of CV and FL to control the IR-HAD process of kiwifruit and monitor its color and shape changes. Two ANN models were used to optimize the fuzzy controller and to predict MC from image information, respectively. The datasets were then divided into three groups (65 % for training, 15 % for validation, and 25 % for testing), and RMSE and R^2 were used to evaluate the model performance. The results showed that this system reduced the drying cycle time by 40 % and significantly improved the appearance properties after drying compared to conventional HAD (Hossein Nadian et al., 2017). In addition, the BP-ANN based on ANN optimization showed excellent prediction performance in the

prediction model of MC in typical fruit and vegetable drying processes, especially when combined with LF-NMR (Sun, Zhang, Mujumdar, & Yang, 2019; Sun, Zhang, & Yang, 2019). To solve the problem of the microwave drying process being difficult to monitor online, Zhu et al. designed a microwave drying online monitoring system based on an Extreme Learning Machine (ELM) in combination with CV. By acquiring the image information of cantaloupe in real time to model its drying process using ANN and ELM, ELM showed superior prediction performance (G. Zhu et al., 2023). In addition, Cetin et al. prepared apple slices using sun-drying, convection-drying, and FD. The total phenol, antioxidant, and ascorbic acid contents were evaluated using algorithms, such as ANN, k-nearest neighbor, random forest, Gaussian process (GP), and support vector regression (SVR), with the best performance of SVR and GP (Cetin & Saglam, 2022). Liu et al. used HSI to monitor changes in soluble solid content, titratable acidity, MC, and hardness during the HAD of red dates. A prediction model combining a CNN, bidirectional short-term and long-term memory (BiLSTM), and squeeze excitation (SE) was developed and compared with PLSR and SVR. The results showed that the model could visualize the spatial and temporal distribution of quality attribute parameters of red dates at different drying stages with better prediction performance (Q. Liu et al., 2025). These studies show that real-time detection of changes in the physiotherapy and biochemical properties of fruits and vegetables can be achieved by combining sensor technology and ML to maximize the retention of the nutritional value of dried fruits and vegetables.

4.2. Meat drying

Meat products, especially proteins, fats, and certain essential amino acids, are major sources of energy and nutrients for humans. Due to their health benefits, the demand for meat products is growing steadily worldwide (Biscarra-Bellio et al., 2023). Therefore, the preservation of meat products is of great importance in addressing the global food crisis. Drying is a common method for meat preservation aimed at reducing MC, extending shelf life, and inhibiting microbial growth. Various drying methods have been used to prepare dried meat, including HAD, VD, FD, and microwave drying (Dincer, 2023). For dried meat, preservation of the original color, nutrients, softer texture, and safe microbial residues are important factors to consider in the drying process. In general, meat usually needs to undergo pretreatment steps, such as marinating and pre-cooking, before drying to improve product quality (Mediani et al., 2022). However, improper pretreatment methods and drying conditions usually lead to a decrease in the quality of the dried meat. The drying process of meat is usually accompanied by complex biochemical reactions, especially those affected by pretreatment, which ultimately affect the color, flavor, and aroma of dried meat. AI is currently widely used in food processing, particularly for analyzing and learning from large amounts of data and monitoring the process in real time. Stawczyk et al. (2004) developed an FL-based control system to optimize the meat drying process (Stawczyk et al., 2004). Muñoz et al. analyzed segmented images of dry cured meats using a CNN to assess intramuscular fat (IMF) content. CNN can achieve high accuracy (0.82 and 0.84) under limited training conditions, although performance may be limited owing to pixel labeling (Munoz et al., 2019). The rapid detection of harmful microorganisms during the drying process, which is one of the main risks affecting the safety of dried meat foods, is of critical importance to the food industry. Lippolis et al. investigated the non-destructive detection of ochratoxin A-producing *Penicillium* strains in dried cured meat using a metal-oxide-semiconductor (MOS)-based *E-nose*, and analyzed it using a discriminant function (DFA) with 73 % recognition rate of the selected. The discrimination function (DFA) was used to analyze 60 dry-cured meats with a recognition rate of 73 % (Lippolis et al., 2016). Poligné et al. proposed a ANN model for modeling the pork dewatering-impregnation-soaking process, which was used to predict mass transfer kinetics and quality changes and optimize the process by mathematically correlating process parameters and quality

characteristics (Poligné et al., 2002). In addition, Deepak et al. investigated the rehydration and thermodynamic and spectral properties of convective drying of chicken breast fillets, developed a low-complexity ANN model with 10 hidden-layer units, and compared it with eight semi-empirical models. The results showed that the model provided better predictions (Deepak et al., 2019). In addition, BP-ANN, NIR-ANN, NIR-MPLS, and GRNN have been used to predict the quality of dried meat products (Pezo et al., 2013; Revilla et al., 2020; N. Zhu et al., 2021). These studies suggest that AI-based bionics (E-nose, CV) and hyperspectral techniques have great potential for applications in the meat drying industry. Moreover, expert systems with real-time updating and powerful learning capabilities are necessary for meat-drying studies that require complex pre-processing. However, there are few related studies, especially online real-time monitoring studies, that combine these technologies.

4.3. Nut drying

Nuts, as nutritious leisure food, are deeply liked by most people. However, because most nuts have hard shells, quality assessment of nuts for key parameters is challenging. On the one hand, nuts are usually irregularly shaped and have uneven moisture distribution, which often results in uneven drying. However, inappropriate drying conditions may affect the taste and color of nuts, negatively affecting consumer preferences. Conventional moisture assessments and sensory attributes of nuts are usually performed using destructive direct testing with a time lag. AI-based predictive models are currently of great interest to researchers. HSI techniques and ML have been used to modeling the moisture distribution in nuts. Farrar et al. used HSI and ML to predict moisture in individual macadamia nuts and compared the effects of imaging data extraction methods (different orientations) on the prediction results. Prediction models were developed using PLSR, ANN, SVM and GRP, with PLSR having higher prediction accuracy for shelled nuts and kernels in the band between 760 and 967 nm, with R^2_T of 0.96 and 0.99, respectively. ANN and GRP, on the other hand, had the highest prediction accuracy for the kernel, with both having R^2_T of 0.99, although this can increase the computational cost (Farrar et al., 2024). In addition, HSI has been used to predict nutrient content changes in nuts. Gama et al. investigated the potential of imaging techniques to predict peroxide values (PV), free fatty acid (FFA) levels, and fatty acid concentrations in Canarium and Macadamia nuts. The results showed that both ANN and PLSR had high prediction accuracy but poor results for PV and linoleic acid prediction (Gama et al., 2024). Walnuts are among the most popular nuts and are subject to MC in their broken shells for deep processing (Jiannan et al., 2022). However, the drying process of walnuts is nonlinear, variable, complex, and difficult to monitor owing to their shells. In particular, the detection of MC usually requires crushing and has a large margin of error. Conventional empirical models are only suitable for prediction under limited conditions and cannot describe the complete model (Martinez-Martinez et al., 2015). Currently, AI has been reported to be used to optimize the drying process of walnuts due to its efficient and accurate learning and prediction capabilities. Yang et al. obtained a model for drying-assisted walnut crushing using GA-ANN (Genetic algorithm-based optimization ANN). By responding to the parameters of drying time, specific energy consumption, high kernel rate, whole kernel rate, and shell crushing rate, the optimal process parameters (including infrared temperature, wind speed, and loading direction) were obtained with a relative error of less than 4 % (Taoqing et al., 2023). In addition, ELM was used to model the fixed-bed drying process of biltmore nuts. The researchers compared the fit of the mathematical model to the ELM model by collecting a large amount of drying data and found that the ELM model can better capture the trend of the drying rate over time and consider the effects of multiple factors (external environment). However, the study found that the prediction ability of ELM in long-term prediction or under extreme conditions needs to be further verified,

and the generalization ability of the model needs to be strengthened (Asim Balbay et al., 2012). These studies suggest that sensor technologies, such as CV and HSI combined with AI, have great advantages and potential for nut-drying process prediction and real-time visualization.

5. Prospects and challenges for AI combined with 5G and IoT in the food drying industry

5.1. Prospects for the future

With the popularization of computers, automation, and PLC technologies, most food drying plants have adopted automated drying equipment such as PLC-controlled HAD, spray drying, and FD. However, although this equipment can realize automatic control and recording of some parameters, it still requires manual monitoring and adjustments. The traditional drying mode can face problems such as uneven drying, unstable quality, and resource wastage. Currently, most of the food-drying industry is still in the 3.0 period of automation. With the emergence of technologies such as fifth-generation (5G) cellular technology, AI, and IoT, the intelligent and connected 4.0 era of food drying factories has become possible. With the dramatic increase in IoT devices, 5G will bring huge benefits, especially in terms of energy optimization (Fig. 2). In addition, 5G provides fast and low-latency links, laying the groundwork for an online real-time monitoring system based on a variety of sensors for application in the food smart drying industry (El Boudani et al., 2020; Meira et al., 2023). This system includes: 1) utilizing the high-bandwidth and low-latency characteristics of 5G technology, it can realize rapid data transmission between the drying equipment and sensors, which is no longer limited by the restrictions of traditional networks, making real-time monitoring possible; 2) various IoT devices can be arranged in various aspects of the drying process to monitor temperature, humidity, pressure, airflow, energy consumption, MC, and other quality characteristic parameters and summarized to a cloud platform through 5G aggregation to the cloud platform; 3) these data are analyzed and processed through AI algorithms to gain insight into the dynamic drying process and provide decision-making services (Fig. 3). This system has the following advantages: 1) it can optimize the AI algorithm through the analysis and integration of historical data to achieve real-time updates of the system; 2) it realizes fully automatic control of the production process, avoiding excessive human intervention, reducing energy consumption, and carbon footprint; 3) it carries out intelligent monitoring of the equipment and predicts potential faults; and 4) it records the entire process in real time, realizing the traceability of the dried products, and guaranteeing food safety. Through the introduction of 5G and IoT technologies, combined with AI, precise control of the food drying process, real-time assessment of the quality of the product, and timely adjustment of the process parameters can be realized, thus ensuring the quality of the final product. In addition, although there are no application cases in the food field yet, the related technology has received much attention in the automobile manufacturing and agriculture fields (Alselek et al., 2022; Zhivkov et al., 2023).

5.2. Challenges and solutions

However, there are still some challenges in the application of these technologies in the food drying industry: 1) Data are scarce and inaccessible: data under extreme environments (low temperature, low pressure, high temperature and high pressure, etc.) are difficult to obtain in real time, and some high-value food materials have high costs and insufficient data. 2) Data labeling challenge: some key indicators (internal moisture content, precise texture properties) require destructive detection, and real-time labeling is difficult. In addition, labeling is subject to subjective influence and human bias. 3) Domain adaptation: the drying conditions and effects of experimental drying equipment are significantly different from those of large-scale factory drying

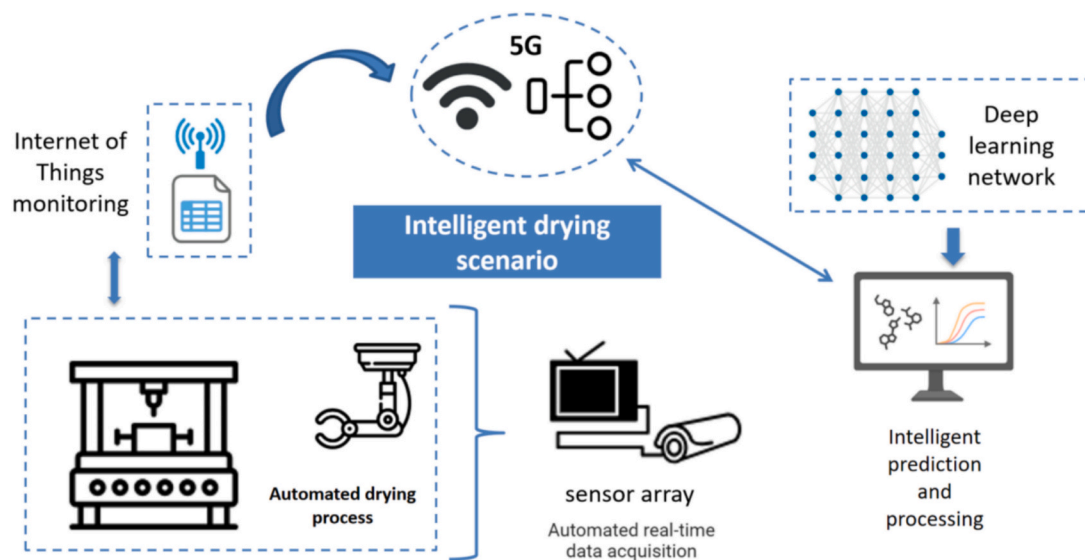


Fig. 3. A typical AI-based online real-time monitoring system.

equipment. There are differences in the drying kinetics of different materials. 4) Model generality: the AI model may have deviations between laboratory conditions and factory conditions due to differences in equipment and environment. 5) Hardware integration: the drying equipment in some factories is old (no built-in chip), and different sensors (most do not support IoT) layout may lead to AI model failure. In addition, extreme environments are difficult to deploy and require high sensor accuracy. 6) Cost constraints: most industrial environments have high network latency, and it costs a lot to arrange 5G networks to meet the millisecond control of the factory. In addition, the current generalization capability of AI is insufficient to support its ability to learn, monitor, and control the entire processing environment. Therefore, the following measures can be taken to address potential industry obstacles: 1) Hardware optimization. High-precision sensors or optimized sensor arrays are adopted to ensure the accuracy and reliability of the data collected in extremely dry environments. 2) Model optimization. Embed the drying dynamics equation in machine learning or deep learning to establish a virtual drying system model to improve the interpretability of AI prediction. 3) Cost control. Develop cloud-based AI services, establish a large-scale database of physical properties of food materials, share training data, improve model efficiency and reduce computing costs.

6. Conclusion

This review describes and analyzes the potential applications of advanced artificial intelligence (AI) modeling in food drying. It also describes the application of AI and advanced sensor technologies for the prediction, control, and optimization modeling of MC, mass and heat transfer, chemical/biochemical reactions, and other quality parameters in food drying processes. In addition, the process prediction and optimization of AI in different types of food drying processes were explored. The results show that ANN, RNN, CNN, AFNIS, and other optimized algorithms, as well as their combined applications, can be effective in helping researchers gain a deeper understanding and explanation of food-drying processes. Finally, the idea of a smart drying system with multisensor online monitoring based on 5G and IoT technologies is proposed to provide suggestions for future food drying research. However, current AI models still have limitations, such as poor interpretability, insufficient generalization capability, and high cost. Therefore, expanding the training dataset (the establishment of a large food database), adaptability to different drying technologies, program simplification, and software/hardware updates (cloud services, integration of big data, integration of intelligent dryers, etc.) may be the focus of future

intelligent drying research.

CRedit authorship contribution statement

Zhe Wang: Writing – original draft, Methodology, Formal analysis, Data curation. **Min Zhang:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Arun S. Mujumdar:** Writing – review & editing, Methodology, Investigation. **Jiacong Lin:** Writing – review & editing, Resources, Investigation. **Dongxing Yu:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

There are no conflicts to declare.

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Data availability

Data will be made available on request.

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